

Eye of Poirot — Methodology v0.2.1

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Methodology — sanction-exposure metrics (v0.2.1, frozen 2026-07-13)

This document is the frozen Phase A specification for *Eye of Poirot*. It is written so that an independent analyst can rebuild every published quantity from the event-level tables. Version 0.2 supersedes the draft W1, D1, W2, W2*, DDM, DDMF, and DF nomenclature. Version 0.2.1 is a pre-registration revision made later the same day, before 2026 M101 and before any registration was submitted; its changes are enumerated in §6 and in `docs/AMENDMENTS.md`.

Analysis scope. FIFA World Cups 2014, 2018, and 2022 in full, plus only completed 2026 matches M1–M100. The unplayed 2026 semi-finals, third-place match, and final (M101–M104) are outside this frozen snapshot.

Third-place match exclusion. Cards shown in a third-place match create no exposure terms, and fouls committed in it enter no denominator. The fixture itself stays in the schedule universe: a suspension actually served in a third-place match still counts as a served match, and the availability components of ω (minutes, squad membership, suspension and injury intervals) include it. The rationale is that the third-place fixture carries no onward qualification stake, so its own events are excluded, while sanctions spilling into it from earlier matches remain real losses.

Analysis cohort. Match exposure is reported for every knockout match. Suspension exposure is reported for every team that entered the knockout stage; full-tournament tables may retain eliminated group-stage teams as an auditable secondary cohort.

0. Design goal and claim boundary

A card is converted to **sanction exposure**, measured in minutes. The match metric prices the remaining in-play time affected by a card. The suspension metric also prices the time for which a caution remains capable of triggering a future ban and the nominal time lost when a ban or dismissal is served.

These quantities are descriptive. They do not identify referee intent, causation, or whether an individual foul deserved a card. Score state, foul severity, field position, player behaviour, tactics, and referee assignment remain uncontrolled unless a later registered amendment says otherwise.

1. Data model and sources

1.1 Match, event, and player tables

For edition y , team i , match m , player p , and card event c :

Symbol	Meaning
t_c	cumulative event minute from the source label; 45'+2' becomes 47
$T_{end,m}$	end of the final in-play period, including stoppage and extra time but excluding a penalty shoot-out
F_{im}	fouls committed by team i in match m
F_i	all-tournament fouls committed by team i within the edition cutoff, excluding the third-place match
M_p	nominal minutes played by player p
T_i	sum of nominal match durations for team i : 90 or 120 per match
S_p	nominal minutes unavailable through a dismissal or a suspension being served
I_p	nominal minutes unavailable through a documented injury or illness

The normalized data model consists of match, card, team-match foul, player- match participation, disciplinary-decision, suspension, availability, and evidence tables. Player and team identifiers are retained whenever a source exposes them; normalized names are display fields, not join keys where an identifier exists.

FIFA period codes distinguish cards shown during play (3, 5, 7, 9), an interval (0), post-play administration (10), and a penalty shoot-out (11). All remain in the card census. Only the in-play class enters E_m .

1.2 Source hierarchy

1. FIFA calendar, timeline/event, team-statistics, and match-centre data;
2. FIFA Full-Time Match Reports and tournament technical reports;
3. national federation or club statements;
4. established match reports (including BBC, ESPN, Reuters, and L'Équipe);
5. FotMob, Transfermarkt, or Wikipedia as a documented fallback or cross-check.

FIFA's archived 2014 timelines do not contain usable final-period markers. For 2014 only, $T_{end,m}$ is reconstructed from archived FotMob **matchDetails** period timestamps. In penalty-shoot-out matches, **gameEnded** is not used because it includes the shoot-out; $T_{end,m}$ is 120 plus the announced second-extra-time added time. If that announcement is absent, 120 is retained as an explicit lower bound rather than imputed.

Complete 2014 team-match foul totals are taken from archived HuffPost World Cup match-statistics pages, which credit Opta as data provider. FotMob has a usable foul statistic for 19 of the 64 matches and is retained as an independent cross-check: 13 match-level home/away pairs agree exactly and six differ by one foul for one team. The complete HuffPost/Opta layer is used in the published denominator; layers are never averaged or silently merged.

Player participation for all four editions is reconstructed from archived FIFA live match-centre payloads using FIFA player identifiers. Stoppage labels are reduced to the nominal clock for participation (for example, 45+4 maps to nominal minute 45) while their cumulative value remains in the event table for E_m .

Raw responses and PDFs are held in the private, gitignored archive. Public derived tables carry source URLs, extraction notes, and the snapshot manifest needed to recollect or audit them.

1.3 Event scope

Card type is one of ordinary yellow (Y), second caution in the same match followed by dismissal (Y2), or direct red (R). Cards to coaches or other team officials are retained in the event census but excluded from all player metrics. A card shown during a penalty shoot-out is outside the in-play match metric and is separately flagged.

1.4 Frozen rule and collection references

- [FIFA: 2026 double reset](#)
- [FIFA Council: 2026 regulations amendment](#)
- [FIFA: Russia 2018 disciplinary reminder](#)
- [FIFA: Qatar 2022 suspension explanation](#)
- [HuffPost 2014 World Cup statistics, data credited to Opta](#)
- FIFA calendar and timeline endpoints are recorded per row in the raw manifest and public derived tables.
- FotMob 2014 fallback endpoint pattern: https://www.fotmob.com/api/data/matchDetails?matchId=<match_id>.

2. Match exposure E_m

2.1 Per-card exposure

For a player card shown during an in-play period:

$$x_m(c; \rho) = \rho_c \max(0, T_{end, m(c)} - t_c), \quad \rho_c = \begin{cases} 1, & c = Y, \\ \rho, & c \in \{Y2, R\}. \end{cases}$$

The primary dismissal multiplier is $\rho = 2$. Sensitivity values are $\rho \in \{1, 1.5, 2\}$. A second-yellow event is represented once as Y2; its earlier yellow remains an ordinary yellow event, so no card event is silently discarded.

2.2 Team-match exposure and differentials

$$E_m(i, m; \rho) = \sum_{c: i(c)=i, m(c)=m} x_m(c; \rho).$$

Match exposure is defined for knockout matches other than the third-place match, which is excluded together with its fouls (§Third-place match exclusion).

For opponent o :

$$\Delta E_m(i, m) = E_m(i, m) - E_m(o, m).$$

Positive values mean that team i carried more remaining-time card exposure than its opponent. The two team perspectives for a match must be exact opposites.

The foul-normalized quantities are

$$e_m(i, m) = \frac{E_m(i, m)}{F_{im}}, \quad \Delta e_m(i, m) = e_m(i, m) - e_m(o, m).$$

For the winner-perspective figures, the same normalized differential is displayed as

$$\Delta E'_m(i, m) = \frac{E_m(i, m)}{F_{im}} - \frac{E_m(o, m)}{F_{om}} = \Delta e_m(i, m),$$

where i is the FIFA-recorded winner. The prime is a reporting label, not an additional division applied to e_m .

If either foul denominator is zero or missing, the corresponding normalized quantity is missing and is never replaced by zero.

3. Suspension exposure E_s

3.1 Reset windows

A **reset window** is a sequence of matches over which single cautions carry forward.

- 2014, 2018, and 2022: one window from the group stage through the quarter-finals; single cautions are cancelled after the quarter-finals.
- 2026: the group stage is one window and R32–R16–quarter-final is a second window; single cautions are cancelled after the group stage and again after the quarter-finals.

This 2026 treatment follows FIFA’s 28 April 2026 rules amendment. A 2026 group card therefore never carries into the knockout stage.

3.2 Knockout-impact scope

X_s prices only the knockout impact of the disciplinary record. Terms are created by:

- cards shown in knockout matches other than the third-place match;
- for 2014, 2018, and 2022 only, a **carried-in caution**: a player entering the knockout stage with exactly one pending single caution from the group stage is treated as if that caution had been shown at minute 0 of the team’s first knockout match. The group card itself creates no term; the carried-in caution replaces it. Because cautions pair chronologically, at most one caution can be pending at knockout entry.

2026 group cautions are cancelled at the group-stage reset and never create a term. Group cards otherwise create no direct terms: in-group risk and a suspension served in a group match are group-stage impact and outside this metric. A suspension earned in the group stage but actually served in a knockout match is knockout impact and enters the served term of §3.4.

3.3 Caution risk intervals

Each in-scope ordinary caution c opens a risk interval $L(c)$ that runs until the earliest of the applicable reset or the player’s next suspension-causing card (a cumulative second caution or any dismissal), measured at minute granularity.

In-match remainder. The default remainder is nominal,

$$r(c) = \max(0, B_c - t_c), \quad B_c = \begin{cases} 90, & \text{regulation period,} \\ 120, & \text{extra time,} \end{cases}$$

with two exact-clock replacements in which the remainder is $\max(0, T_{\text{end}, m(c)} - t_c)$ instead:

1. a caution received in stoppage time or in extra time — its remaining threat runs on the observed clock, so a stoppage-time caution is not priced at zero;
2. any caution shown in a semi-final or a final — after the quarter-final reset no future fixture can be reached by accumulation, and while the semi-final is being played neither side knows who will advance, so the live threat is a same-match second caution costing the next fixture, and that threat runs on the observed clock for the exact remaining minutes.

Horizon blocks. Untriggered, the interval extends past the current match by $90N(c)$ fixture blocks, where $N(c)$ counts the team’s potential fixtures strictly after the card and before the applicable reset, following the full bracket path and not shortened by the team’s actual elimination. Semi-final and final cautions have $N = 0$ and use the exact-clock remainder above.

Edition	R32	R16	QF	SF/final
2014/2018/2022	—	1	0	exact clock
2026	2	1	0	exact clock

A carried-in caution at minute 0 of the first knockout match prices $90 + 90N$ of that match, so 180 minutes at a 2014–2022 round of 16.

Stop rule (minute granularity). If the interval is stopped by a suspension-causing card at t_2 :

- same match: $L(c) = \max(0, \min(t_2, K_c) - t_c)$, where K_c is the applicable in-match cap — B_c for the nominal case, $T_{end, m(c)}$ for the two exact-clock cases;
- later match (nominal-remainder cautions only): $L(c) = r(c) + 90G + \min(t_2, 90)$, where G counts the team's fixtures strictly between the two matches.

The triggering card itself receives no forward blocks: a same-match **Y2** keeps only its dismissal term, and a cross-match triggering caution keeps its own in-match remainder. A stopped interval never exceeds its untriggered value.

3.4 Player-level sanction minutes

For player p , let C_p^Y be in-scope ordinary cautions (§3.2, a carried-in caution included), $L(c)$ the §3.3 risk interval, C_p^D be in-scope dismissal events (**Y2** or **R**), and D_p , A_p be the numbers of dismissal and two-caution suspension matches actually served **in knockout matches** — the third-place match included — within the observed edition, whether the suspension was earned in the group stage or the knockout stage. Then

$$X_s(p; \rho, \mu) = \sum_{c \in C_p^Y} L(c) + \sum_{c \in C_p^D} \rho \max(0, T_{end, m(c)} - t_c) + 90\mu (D_p + A_p).$$

The primary served-suspension multiplier is $\mu = 1.25$, with sensitivity $\mu \in \{1, 1.25, 1.5\}$. Suspension terms are included only when the loss is observed within the edition cutoff. Pending sanctions after the team's final observed match are recorded but do not create fictitious served minutes. An automatic sanction is initially mapped to the next team match. A sourced disciplinary decision may enumerate additional service matches, carry a ban into the tournament, or defer execution. Only the enumerated matches enter $D_p + A_p$, and every claimed served match must also be confirmed by the official lineup. A deferred decision is published as **deferred** and creates no served term.

3.5 Player opportunity weight ω

$$\omega_p = \frac{M_p}{T_i - |S_p \cup I_p|}.$$

All components use nominal match minutes. **S** and **I** are interval sets and are unioned before subtraction so overlapping reasons cannot be counted twice. The denominator includes only matches in which the player belonged to the tournament squad. Values must satisfy $0 \leq \omega_p \leq 1$; a violation is a data error, not a value to clip silently.

I_p includes a documented in-match injury remainder and documented whole-match absences. A substitution alone is not injury evidence. Every positive injury interval must have a source URL and an evidence note. An unexplained absence stays in the denominator. **S_p** includes the in-match remainder after a dismissal and a suspension actually served.

3.6 Team exposure and normalization

$$E_s(i; \rho, \mu) = \sum_{p \in i} \omega_p X_s(p; \rho, \mu), \quad e_s(i; \rho, \mu) = \frac{E_s(i; \rho, \mu)}{F_i}.$$

The primary denominator is all fouls committed by the team across group and knockout matches within the edition cutoff, excluding the third-place match. A prespecified sensitivity uses knockout-stage fouls only, likewise excluding the third-place match. The edition pooled ratio is

$$\bar{e}_s = \frac{\sum_i E_s(i)}{\sum_i F_i},$$

not the unweighted mean of team ratios.

4. Availability, suspension, and evidence coding

Availability is coded at player $\times$ team-match grain with one of: **played**, **bench**, **suspended**, **injured**, or **unexplained**. Only players with positive unweighted sanction minutes $X_s > 0$ require complete availability coding for the primary build.

1. Participation and bench status come from official line-ups or the best archived match record.
2. Automatic suspensions are derived from the chronological card ledger; sourced disciplinary decisions may extend, carry in, or defer them. Every claimed service match is checked against its official line-up.
3. Injury/illness requires explicit evidence. A commentary phrase such as “replaced because of an injury” may establish the in-match remainder; a whole-match absence requires an independent team or media report.
4. If evidence cannot distinguish tactical non-selection from injury, status is **unexplained** and injury minutes are zero.

If a rule-derived suspension trigger conflicts with the next official lineup, both observations are published with status **conflict**; the event remains in the card ledger, but no served-suspension term or unavailable interval is inferred. This implements the requirement that D_p and A_p count only losses observed as served.

Each positive injury row contains **player**, **match**, **status**, **interval**, **source_url**, evidence tier, and a short quotation-free paraphrase. The evidence audit is published in `data/sources/injury-evidence.md`.

5. Prespecified sensitivity and quality control

The full grid is

$$\rho \in \{1, 1.5, 2\}, \quad \mu \in \{1, 1.25, 1.5\},$$

crossed with all-tournament versus knockout-only foul denominators. The primary setting is $(\rho, \mu) = (2, 1.25)$ with all-tournament fouls.

A boundary sensitivity replaces $T_{end,m}$ by $T_{end,m} - 1$ in every component that uses the observed final-whistle clock: $x_m(c; \rho)$, the dismissal term in X_s , and the exact-clock caution remainders of §3.3 (stoppage/extra-time receipt and semi-final/final cautions). Nominal caution remainders $r(c)$ and served-match blocks are unchanged. This checks inclusive-versus-exclusive end-minute coding and is not a second primary specification.

5.1 Prespecified descriptive depth check

After each edition’s full computation, the association between tournament depth and e_s is reported as a prespecified descriptive confound check:

- unit: knockout-stage teams of one edition; depth = number of matches played within the edition cutoff, excluding the third-place match (so a losing semi-finalist and a finalist differ);
- statistic: Kendall τ_b between depth and e_s at the primary setting, with a within-edition permutation p (10,000 permutations, two-sided, fixed seed 20260713);
- editions are reported separately and never pooled: cross-edition strength structures differ too much for a pooled coefficient to be meaningful;
- a prespecified secondary depth proxy repeats the test with the World Football Elo pre-tournament rating (eloratings.net) in place of matches played, once that table is collected and archived;
- claim boundary: this is a descriptive check of whether e_s drifts mechanically with tournament depth. Depth is partly endogenous to discipline itself (suspensions weaken teams), so the coefficient is not an estimate of a causal strength effect and is not reported as one.

Required invariants:

- player-card census: 2014 = 194, 2018 = 224, 2022 = 228, 2026 M1–M100 = 270;
- $E_m \geq 0$, $E_s \geq 0$, and $0 \leq \omega \leq 1$;
- the stop rule never yields an interval longer than the untriggered interval;
- group-stage and third-place cards create no exposure terms, and served suspensions are counted only when the service match is a knockout match;
- two match perspectives satisfy $\Delta E_m(i, m) = -\Delta E_m(o, m)$, and likewise for Δe_m when defined;
- every positive injury interval has a non-empty source URL;
- sourced disciplinary decisions retain their decision URL and lineup check;
- 2026 inputs and outputs contain no event from M101–M104.

Hand-checked fixtures cover regulation stoppage, extra time, shoot-outs, direct red, same-match second caution, cross-match accumulation, a reset, and an injury-adjusted opportunity denominator in every edition where applicable. Four

matches—one per edition—also have an independent ESPN/Opta match-page check; source-display differences remain visible as **AUDIT**, not forced agreements.

6. Frozen disclosure and amendments

The analyst had seen earlier exploratory rankings and Argentina-focused draft figures for 2018, 2022, and 2026 before freezing v0.2. Those drafts used the superseded W1/D1/W2/W2* family and included two known rule errors: a semi-final accumulation block after the quarter-final reset, and 2026 group cautions carrying into the knockout stage. They are not confirmatory outputs.

This document, the input manifest, and the analysis cutoff were frozen on 2026-07-13 before 2026 M101. Any later change to a definition, inclusion rule, source hierarchy, parameter, or test must be dated in `docs/AMENDMENTS.md` with its reason and its expected directional effect. Corrections to raw transcription that do not change a rule are logged as data errata. No silent back-editing is permitted after registration.

v0.2.1 pre-registration revision (2026-07-13, before 2026 M101 and before any registration). Review against the analyst’s original commissioned definitions found that v0.2 had formalized four rules differently from the commission, and one agreed rule was added. All five changes were applied before the OSF submission; no confirmatory analysis had been run under either variant. The changes and their expected directional effects on X_s :

1. knockout-impact scope replaces whole-tournament pricing (group cards now enter only as a 2014–2022 carried-in caution; group-served suspensions no longer count) — decreases X_s for players carded in the group stage;
2. the second-caution stop rule is applied at minute granularity instead of whole-fixture granularity — decreases stopped intervals;
3. cautions received in stoppage time or extra time use the exact-clock remainder — increases those intervals (a stoppage caution is no longer floored at zero);
4. semi-final and final cautions use the exact-clock remainder — increases those intervals slightly relative to a nominal remainder;
5. the third-place match is excluded at the event level (no exposure terms, no denominator fouls; service and availability there still count) — removes small amounts from both numerator and denominator.

A prespecified depth check (§5.1) was added at the same time, before any correlation had been computed. No registration existed under the v0.2 rules.

7. Symbol and output migration

Draft label	v0.2 label / disposition
D1	E_m , exact match exposure
D1 difference	ΔE_m
D1/foul difference	Δe_m
W1	$r(c)$, retained only as a nominal within-match component
W2, W2*, adjusted W2*	superseded by X_s , E_s , and e_s
DDM, DDMF, DF	retired; use <code>sanction_exposure</code> and the symbols above

Canonical machine-readable columns are `exp_match_min`, `exp_susp_min`, `d_exp_match`, `exp_match_per_foul`, `exp_susp_per_foul`, and `omega`. Official figure filenames spell out `match-exposure` or `suspension-exposure`; draft abbreviations do not appear in final figures.

Version history

- **v0.2.1 (2026-07-13, pre-registration).** Knockout-impact scope with 2014–2022 carried-in cautions; minute-granular stop rule; exact-clock remainders for stoppage/extra-time and semi-final/final cautions; event-level third-place exclusion; knockout-only served-suspension counting; prespecified depth check (§5.1); boundary sensitivity extended to all observed-clock components. Enumerated with directions in §6.
- **v0.2 (2026-07-13).** Terminology reset to sanction exposure; unified end-clock convention; formal 2026 two-window reset; second-caution stop rule; primary ρ , μ , denominator, evidence, sensitivity, and amendment rules frozen; scope extended to 2014 and capped at 2026 M100.

- **v0.1 (2026-07-12).** Exploratory draft. Superseded and retained only for provenance through the archived draft figures.